PROJECT :: Managed Memory and Multiprocessing in a Faux File Hasher

Start Assignment

- Due Thursday by 11:59pm
- Points 100
- Submitting a file upload
- File Types c
- Available until Nov 11 at 11:59pm

First look at the prompt.
Before you start coding, read the **full assignment prompt carefully (https://csus.instructure.com/courses/21989/pages/project-os-managed-memory-and-multiprocessing-in-a-faux-file-hasher)** . This includes all sections, examples, and notes. Every detail matters. The provided description explains how the project builds across multiple steps and how the grading process works. You are expected to understand what each part of the assignment requires before you begin coding.

You will **download the provided scaffold file,** `hashproj.c` **(https://csus.instructure.com/courses/137456/files/26156153?wrap=1)** ↓ **(https://csus.instructure.com/courses/137456/files/26156153/download?download_frd=1)** . This is the file you will edit and submit. Do not rename it, and do not submit multiple files — all of your work should occur in the clearly marked *STUDENT SECTION* of `hashproj.c` . The provided scaffold includes complete infrastructure and testing utilities, so make sure you read through it carefully before making any edits.

Be sure to **read every instruction and code comment thoroughly**. Much of what you need to do — and what *not* to do — is explained in the scaffold itself. Take your time understanding the provided code before writing anything.

Assignment Revision History

The assignment can change. If a newer version is released, it will be clearly labeled (e.g., V002, V003, etc.). Always confirm that you are working from the most recent version before submitting your work. Version changes may include fixes, clarifications, or minor updates to testing details.

Current Revision:

- **V001 : Expect bugs**

Assignment Outline

1. **Review the Full Prompt**
   Read the overview and all steps in the project description. The assignment has three major phases, each building on the previous one. Make sure you understand what each step requires.

2. **Download and Inspect the Scaffold**
   - Download `hashproj.c` from Canvas.
   - Read through the entire file before editing.
   - Do not modify anything inside the `#ifndef GRADING_MODE` section.
   - Only edit functions in the *STUDENT SECTION* as described in the assignment.

3. **Implement Step 1: Baseline Functionality**
   Begin by confirming the scaffold compiles cleanly. Implement the required functions using the system allocator (`malloc` / `free`) to establish a working baseline.

4. **Implement Step 2: Custom Allocator**
   Replace your baseline allocator with your own **First-Fit** memory manager using `mmap()`. This step requires careful debugging and attention to pointer arithmetic.

5. **Implement Step 3: Multi-Process Execution**
   Extend the program to use multiple processes and pipes for communication, ensuring your output remains identical to the single-process version.

6. **Test Frequently**
   Compile and test after each change using the provided commands in the instructions. Use small test files to verify correctness and consistency.

7. **Validate and Prepare for Submission**
   - Remove all debug print statements before submitting.
   - Ensure your program compiles cleanly under Linux with `gcc -Wall -Wextra`.
   - Verify that your output exactly matches the expected format.